A+ Core 1 Memorization Sheet

Print this and keep it beside you. Core 1 is a recall exam: port numbers, wireless standards, cable categories, connector names, RAM and drive facts, the laser printing steps, cloud vocabulary, and which symptom points to which part. Every line here is fair game.

01 The exam itself

FACT	VALUE
Exam	CompTIA A+ Core 1, 220-1201 (A+ needs Core 1 and Core 2, 220-1202)
Questions and time	Up to 90 questions in 90 minutes; multiple choice plus performance-based items
Passing	675 on a scale of 100 to 900 (Core 2 needs 700)
Hardware and Network Troubleshooting	28%
Hardware	25%
Networking	23%
Mobile Devices	13%
Virtualization and Cloud Computing	11%

> Troubleshooting and hardware are more than half the exam. For every part: what it is called, its number (speed, pins, volts, distance), how it fails, and what you check first.

02 Ports and protocols

PORT	PROTOCOL	TRANSPORT	PURPOSE
20, 21	**FTP**	**TCP**	File transfer; 21 commands, 20 data; clear text
22	**SSH**	**TCP**	Secure remote shell; also **SFTP** and **SCP**
23	Telnet	**TCP**	Clear-text remote terminal; replaced by **SSH**
25	**SMTP**	**TCP**	Sending mail between servers (587 for client submission with **TLS**)
53	**DNS**	**UDP** and **TCP**	Name to address lookups (**TCP** for large answers and zone transfers)
67, 68	**DHCP**	**UDP**	Automatic addressing; server listens on 67, client on 68
80	**HTTP**	**TCP**	Web, unencrypted
110	**POP3**	**TCP**	Download mail to one device (995 secure)
137 to 139	NetBIOS / NetBT	**UDP** 137, 138; **TCP** 139	Legacy Windows name service and file sharing
143	**IMAP**	**TCP**	Mail kept on the server, synced across devices (993 secure)
161, 162	**SNMP**	**UDP**	Device monitoring; 162 traps
389	**LDAP**	**TCP**	Directory queries (636 **LDAPS**)
443	**HTTPS**	**TCP**	Web over **TLS**
445	**SMB** / **CIFS**	**TCP**	Windows file and printer sharing
3389	**RDP**	**TCP**	Remote Desktop
123	**NTP**	**UDP**	Time synchronization

PORT	PROTOCOL	TRANSPORT	PURPOSE
514	**Syslog**	**UDP**	Log collection
5900	**VNC**	**TCP**	Cross-platform screen sharing

TCP versus UDP

- **TCP**: connection-oriented (three-way handshake), reliable, ordered, acknowledged, retransmits; web, mail, file transfer, remote desktop
- **UDP**: connectionless, no acknowledgment, low overhead; **DNS** lookups, **DHCP**, streaming, **VoIP**, **SNMP**, syslog, **TFTP**
- A port identifies the service at an address; well-known ports are 0 to 1023; a firewall filters by address and port

03 Wireless

STANDARD	WI-FI NAME	BAND	TOP SPEED	NOTES
802.11a		5 **GHz**	54 Mb/s	Old; short range
802.11b		2.4 **GHz**	11 Mb/s	Oldest common; **DSSS**
802.11g		2.4 **GHz**	54 Mb/s	Backward compatible with b
802.11n	**Wi-Fi** 4	2.4 and 5 **GHz**	600 Mb/s	**MIMO**, 40 **MHz** channels
802.11ac	**Wi-Fi** 5	5 **GHz** only	About 6.9 Gb/s	**MU-MIMO**, 80 and 160 **MHz** channels
802.11ax	**Wi-Fi** 6 and 6E	2.4 and 5 **GHz**; 6E adds 6 **GHz**	About 9.6 Gb/s	**OFDMA**, better in crowds
802.11be	**Wi-Fi** 7	2.4, 5, and 6 **GHz**	About 46 Gb/s	320 **MHz** channels, multi-link

BAND	RANGE AND PENETRATION	CHANNELS	CROWDING
2.4 GHz	Longest range, best through walls	1 to 11 in the **US**; only 1, 6, and 11 do not overlap at 20 **MHz**	Microwaves, Bluetooth, cordless phones, neighbors
5 GHz	Shorter range	Many non-overlapping channels; some require **DFS** (radar avoidance)	Less crowded; 20, 40, 80, 160 **MHz** widths
6 GHz	Shortest range	Widest clean spectrum; **Wi-Fi** 6E and 7 only	Newest devices only

Other radios

- Channel width: wider channels carry more data but overlap more; regulations set which channels and power levels a country allows
- Bluetooth: 2.4 **GHz** personal area network; pairing with a **PIN**; class 2 about 10 m; low energy for wearables
- **NFC**: 13.56 **MHz**, about 4 cm, tap to pay and pair; **RFID**: tags read by radio at a distance, passive (no battery) or active
- Long-range fixed wireless and cellular (**3G**, **4G LTE**, **5G**) serve as internet connections where cable and fiber are absent

04 Network hardware and services

DEVICE	WHAT IT DOES	REMEMBER
Router	Connects networks; forwards by **IP** address; the default gateway	Layer 3; **NAT** and **DHCP** in **SOHO** routers
Switch	Connects devices in one network; forwards by **MAC** address per port	Layer 2; managed (**VLANs**, **PoE** settings, monitoring) versus unmanaged (plug and play)
Access point	Bridges wireless clients to the wired network	Not a router by itself; **SSID**, channel, security

DEVICE	WHAT IT DOES	REMEMBER
Patch panel	Terminates building cable runs at the closet; patch cables to switch ports	Punched down with a 110 tool; labeling
Firewall	Allows or blocks traffic by address, port, and rule	Stateful; hardware or software
PoE injector / PoE switch	Sends power over the data cable to **APs**, phones, cameras	**802.3af** 15.4 W; **802.3at PoE+** 30 W; **802.3bt PoE++** 60 W (Type 3) and 100 W (Type 4)
Cable modem	Internet over the cable **TV** coax (**DOCSIS**)	F-type connector; shared neighborhood bandwidth
DSL modem	Internet over the phone line	**RJ11**; distance to the exchange limits speed
ONT	Optical network terminal; converts fiber to Ethernet at the premises	Fiber to the home
NIC	Network interface card; wired or wireless	Burned-in 48-bit **MAC** address, six hex pairs, first half is the manufacturer
Hub	Obsolete repeater that sends every frame to every port	Collisions; replaced by switches

SERVER ROLE OR APPLIANCE	PROVIDES
DNS	Names to addresses
DHCP	Addresses, masks, gateways, **DNS** servers by lease
File share and print server	Shared folders (**SMB**) and shared printers with queues
Mail server	**SMTP** out, **IMAP** or **POP3** in
Syslog	Collects logs from devices
Web server	**HTTP** and **HTTPS**
AAA	Authentication, authorization, accounting (**RADIUS**, **TACACS+**)
Database server	**SQL** data for applications
NTP	Time
Spam gateway	Filters inbound mail before the mail server
UTM	Unified threat management: firewall, **IDS**/**IPS**, antivirus, content filter in one box
Load balancer	Spreads requests across several servers; high availability
Proxy server	Fetches web content on behalf of clients; caching, filtering, logging
SCADA and legacy embedded	Industrial control; old, fragile, isolate on their own network
IoT	Smart devices: thermostats, cameras, speakers, locks; segregate and patch

05 IP addressing and configuration

ITEM	VALUE
IPv4 address	32 bits, four octets 0 to 255, such as 192.168.1.20
Private ranges	10.0.0.0 to 10.255.255.255 (/8); 172.16.0.0 to 172.31.255.255 (/12); 192.168.0.0 to 192.168.255.255 (/16)
Public addresses	Everything else; routable on the internet; shared through **NAT** at the router
APIPA	169.254.x.x: the client got no **DHCP** answer; link or **DHCP** server problem
Loopback	127.0.0.1 (**IPv6** ::1): tests the local **TCP**/**IP** stack

ITEM	VALUE
Subnet mask	255.255.255.0 = /24 (254 hosts); 255.255.0.0 = /16; 255.0.0.0 = /8; same network portion means direct delivery, otherwise via the gateway
Default gateway	The router's address on the local network; must be in the same subnet as the host
Static versus dynamic	Static typed in (servers, printers); dynamic from **DHCP** (workstations, phones); reservations give a fixed address by **MAC**
IPv6 address	128 bits, eight hex groups; :: compresses zeros; fe80:: link-local on every interface; 2000::/3 global; no **NAT** needed

DNS RECORD	PURPOSE
A	Name to **IPv4** address
AAAA	Name to **IPv6** address
CNAME	Alias of another name
MX	Mail server for a domain
TXT	Free text; carries **SPF**, **DKIM**, and **DMARC**
SPF (TXT)	Which servers may send mail for the domain
DKIM (TXT)	Cryptographic signature proving the message was not altered and came from the domain
DMARC (TXT)	Policy telling receivers what to do when **SPF** or **DKIM** fails (none, quarantine, reject) and where to send reports

DHCP, VLAN, VPN

- **DHCP** scope: the pool of addresses; lease: how long a client keeps one; reservation: a fixed address tied to a **MAC**; exclusion: addresses in the range never handed out
- **DORA**: Discover, Offer, Request, Acknowledge
- **VLAN**: a logical network on a managed switch; separates traffic (voice, guest, devices) without separate hardware; a router connects **VLANs**
- **VPN**: an encrypted tunnel across a public network; site-to-site or client; the client appears to be on the office network

06 Internet connection and network types

CONNECTION	MEDIUM	TRAITS
Fiber	Glass, **ONT** at the premises	Fastest, symmetric, immune to interference
Cable	Coax, **DOCSIS** modem	Fast, shared with neighbors, asymmetric
DSL	Copper phone line	Distance-limited, slower; always-on
Satellite	Dish to orbit	Available anywhere; high latency; weather affects it; data caps
Cellular	**4G LTE**, **5G**	Mobile; hotspot and tethering; caps
WISP	Fixed wireless from a tower to an antenna	Rural; line of sight

NETWORK TYPE	SCOPE
PAN	Personal: Bluetooth, a few meters
LAN	One building or site
WLAN	A **LAN** over **Wi-Fi**
MAN	A city or campus
WAN	Between sites, across regions; the internet
SAN	Storage area network: block storage over a dedicated network (Fibre Channel, **iSCSI**)

07 Cables and connectors

COPPER CATEGORY	SPEED	BANDWIDTH	NOTES
Cat 5	100 Mb/s	100 **MHz**	Obsolete
Cat 5e	1 Gb/s	100 **MHz**	Minimum for gigabit
Cat 6	1 Gb/s (10 Gb/s to 55 m)	250 **MHz**	Tighter twists, spline
Cat 6a	10 Gb/s to 100 m	500 **MHz**	Common for 10 gigabit
Cat 7	10 Gb/s	600 **MHz**	Shielded; **GG45** or **RJ45**
Cat 8	25 or 40 Gb/s to 30 m	2,000 **MHz**	Data center runs

Copper rules

- Maximum run 100 m (328 ft) including patch cables; **RJ45** eight-pin plug; **RJ11** four or six-pin phone plug
- **T568A** order: white-green, green, white-orange, blue, white-blue, orange, white-brown, brown. **T568B**: white-orange, orange, white-green, blue, white-blue, green, white-brown, brown
- Same standard both ends = straight-through (device to switch); A on one end and B on the other = crossover (like devices; rarely needed with auto-**MDIX**)
- **UTP** unshielded (normal); **STP** shielded for high-interference areas; plenum-rated jackets for air-handling spaces (low smoke); direct burial for outdoors
- Coax: **RG-6** for cable **TV** and modems, F-type screw connector; **RG-59** for short **CCTV** runs; **BNC** bayonet connector

FIBER	CORE	SOURCE	DISTANCE	CONNECTORS
Single-mode	About 9 microns	Laser	Kilometers	**ST** (bayonet twist), **SC** (square push-pull), **LC** (small clip, most common)
Multimode	50 or 62.5 microns	**LED** or **VCSEL**	Hundreds of meters (**OM3** about 300 m at 10 Gb/s)	Same connectors; orange or aqua jackets

INTERFACE	SPEED	NOTES
USB 2.0	480 Mb/s	5 m cable; black port; Type A, B, mini, micro
USB 3.0 (3.2 Gen 1)	5 Gb/s	3 m; blue port; SuperSpeed
USB 3.1 Gen 2	10 Gb/s	Teal or red port
USB 3.2 Gen 2x2	20 Gb/s	**USB-C** only
USB4	40 Gb/s (80 in version 2)	**USB-C**; tunnels Thunderbolt, DisplayPort, PCIe
Thunderbolt 3 and 4	40 Gb/s	**USB-C** connector; daisy chain; power delivery; Thunderbolt 5 is 80 Gb/s
USB-C	Connector, not a speed	Reversible 24-pin; can carry **USB**, DisplayPort alt mode, Thunderbolt, and up to 240 W power delivery
Lightning	**USB** 2.0 speeds	Apple 8-pin reversible; replaced by **USB-C** on newer devices
Serial RS-232	Kilobits	**DB9** connector; console ports on network gear
SATA	1.5, 3, 6 Gb/s (**SATA** I, II, III)	7-pin data, 15-pin power; 1 m cable; **eSATA** external to 2 m
Molex	Power only	4-pin 5 V and 12 V for older drives and fans

VIDEO	SIGNAL	NOTES
VGA	Analog	15-pin **DE-15**, blue; degrades with length; no audio
DVI	Digital (**DVI-D**), analog (**DVI-A**), both (**DVI-I**)	Single link 1920 x 1200; dual link 2560 x 1600; no audio
HDMI	Digital video and audio	19-pin; Type A full, C mini, D micro; **HDMI** 2.1 up to 48 Gb/s, 8K; **CEC** control
DisplayPort	Digital video and audio	20-pin with a latch; multi-stream daisy chaining; **DP** 2.0 up to 80 Gb/s; mini DisplayPort
USB-C DisplayPort alt mode	Digital	One cable for video, data, and power

08 Displays

TYPE	TRAITS
LCD TN (twisted nematic)	Cheapest, fastest response, poor viewing angles and color
LCD IPS (in-plane switching)	Best color and viewing angles; slower and pricier; professional and most laptops
LCD VA (vertical alignment)	High contrast, deep blacks; middle on angles and speed
OLED	Each pixel emits its own light; perfect blacks, thin, flexible; risk of burn-in; used in phones and premium laptops
Mini-LED	An **LCD** with thousands of tiny backlight zones; high brightness and contrast without burn-in
Inverter	Converts **DC** to **AC** for **CCFL** backlights in old **LCDs**; a dim screen visible with a flashlight means backlight or inverter
Touch screen and digitizer	The digitizer layer senses touch or pen; a cracked digitizer works badly even if the image is fine

Attributes

- Resolution: 1920 x 1080 Full **HD**; 2560 x 1440 **QHD**; 3840 x 2160 4K **UHD**; the native resolution is the only one that looks sharp
- Pixel density in pixels per inch: the same resolution on a smaller panel is sharper
- Refresh rate in hertz: 60 standard, 120 to 240 for gaming; the cable and port must support the rate at the resolution
- Color gamut: sRGB (web), Adobe **RGB** and **DCI-P3** (print and video); a wider gamut shows more colors

09 Memory, storage, and RAID

RAM	PINS (DIMM / SODIMM)	VOLTAGE	TYPICAL SPEEDS	NOTES
DDR3	240 / 204	1.5 V (1.35 V low voltage)	800 to 2133 **MT/s**	Keyed differently from **DDR4**
DDR4	288 / 260	1.2 V	2133 to 3200 **MT/s**	Most common in service now
DDR5	288 / 262	1.1 V	4800 **MT/s** and up	Different key from **DDR4**; on-module power management; two channels per module

Memory rules

- Generations are not interchangeable; the notch position enforces it. **SODIMM** for laptops and small form factors; **DIMM** for desktops
- **ECC RAM** detects and corrects single-bit errors; servers and workstations; the board and **CPU** must support it. Non-**ECC** for consumer systems

- Channels: install matched modules in the color-coded paired slots for dual (or quad) channel bandwidth; a single module runs single channel
- Virtual memory (paging file) uses disk when **RAM** is full; constant paging means add **RAM**

STORAGE	FACTS
HDD	Spinning platters; 5,400, 7,200, 10,000, 15,000 rpm; 2.5-inch (laptop) and 3.5-inch (desktop); slow, cheap per **TB**; fails with clicking and grinding
SSD SATA	No moving parts; about 550 **MB**/s limited by **SATA** III; 2.5-inch or **mSATA** or **M.2 SATA**
SSD NVMe	Talks PCIe directly; **M.2** or add-in card; thousands of **MB**/s (Gen 3 about 3.5 **GB**/s, Gen 4 about 7 **GB**/s)
M.2	A slot and form factor, not an interface: keys B, M, or **B+M**; sizes 2242, 2260, 2280 (22 mm wide, length in mm); can be **SATA** or **NVMe**
SAS	Serial Attached **SCSI**; enterprise drives, 12 Gb/s, dual ports; **SAS** controllers accept **SATA** drives, not the reverse
Removable	**USB** flash drives; **SD**, **microSD**, CompactFlash memory cards; speed classes on the card
Optical	**CD** 700 **MB**; **DVD** 4.7 **GB** (8.5 dual layer); Blu-ray 25 **GB** (50 dual layer)

RAID	HOW	MINIMUM DRIVES	SURVIVES	USABLE CAPACITY
0	Striping	2	No drive failure	100% (sum of drives)
1	Mirroring	2	One drive	50%
5	Striping with distributed parity	3	One drive	All but one drive
6	Striping with double parity	4	Two drives	All but two drives
10 (1+0)	Stripe of mirrors	4	One drive per mirrored pair	50%

10 Motherboards, CPUs, firmware, and power

FORM FACTOR	SIZE	NOTES
ATX	12 x 9.6 in (305 x 244 mm)	Full size; most expansion slots
microATX	9.6 x 9.6 in	Fits **ATX** cases; fewer slots
Mini-ITX	6.7 x 6.7 in (170 mm)	Small builds; one expansion slot

CONNECTOR	FACTS
PCIe	Lanes x1, x4, x8, x16; a card fits in a longer slot; per lane about 1 **GB**/s (Gen 3), 2 **GB**/s (Gen 4), 4 **GB**/s (Gen 5) each direction; graphics cards use x16
PCI	Legacy 32-bit parallel slot; 133 **MB**/s shared
Main power	24-pin (20+4) **ATX**; 4- or 8-pin **EPS** for the **CPU**; 6-pin (75 W) and 8-pin (150 W) PCIe for graphics; **12VHPWR** 16-pin up to 600 W
Drive power	**SATA** 15-pin; Molex 4-pin
SATA and eSATA	7-pin data ports on the board; **eSATA** on the rear panel
Headers	Front panel (power, reset, **LEDs**), **USB** 2.0 and 3.0, audio, fan (3-pin voltage, 4-pin **PWM**), **RGB**
M.2	Key and length printed by the slot; some slots are **SATA** only, some **NVMe** only, some both
CPU sockets	Intel **LGA** (pins in the socket, such as **LGA** 1700 and 1851); **AMD AM4 PGA** (pins on the chip) and **AM5 LGA**; board chipset must match the **CPU** generation; servers may be multisocket

BIOS / UEFI SETTING	WHY IT MATTERS
UEFI versus legacy BIOS	**UEFI**: **GPT** disks over 2 **TB**, Secure Boot, graphical setup, network boot; legacy uses **MBR**
Boot options	Order of devices; **USB** or network boot for installs; disable unused to harden
USB permissions	Disable ports or block storage devices to prevent data theft
TPM	Trusted Platform Module: stores keys for BitLocker and Windows Hello; **TPM** 2.0 required by Windows 11; can be firmware (fTPM) or a chip
Secure Boot	Only signed boot loaders run; blocks boot-sector malware; may need disabling for some Linux or older media
Passwords	Supervisor (setup) versus user (boot) password; cleared by the **CMOS** jumper or battery removal
Fans and temperature	Fan curves, temperature monitoring, thermal shutdown thresholds
Virtualization support	Intel **VT**-x or **AMD-V** must be enabled for hypervisors
HSM	Hardware security module: a dedicated external device for key storage and crypto in servers and enterprises
CMOS battery	**CR2032** coin cell; a dead one resets time, date, and settings at every boot

CPUs and cooling

- x86 is 32-bit, x64 is 64-bit (more than 4 **GB** of **RAM**); **ARM** is a low-power **RISC** design in phones, tablets, Apple silicon, and some laptops; software must be built for the architecture
- Cores run separate work at once; simultaneous multithreading (Hyper-Threading) shows two threads per core; cache L1, L2, L3 speeds up access; clock in **GHz**
- Expansion cards: video (**GPU**, **VRAM**), sound, capture (recording **HDMI** input), **NIC** (wired or wireless); need a free slot, power, and a driver
- Cooling: case fans (intake front, exhaust rear), **CPU** heat sink and fan, thermal paste or pad between **CPU** and sink (replace when reseating), liquid all-in-one coolers; dust is the enemy

POWER SUPPLY	FACTS
Input	110 to 120 **VAC** (North America, 60 **Hz**) or 220 to 240 **VAC** (much of the world, 50 **Hz**); auto-switching or a red voltage selector; wrong setting destroys the supply
Output rails	3.3 V (orange), 5 V (red), 12 V (yellow), ground (black); 12 V feeds **CPU**, **GPU**, and drives
Connectors	24-pin (20+4) main, **EPS CPU**, PCIe, **SATA** power, Molex
Wattage	Add the components' draw and leave headroom; a **GPU** may need 300 W or more by itself
Efficiency	80 Plus Bronze, Silver, Gold, Platinum, Titanium; higher wastes less as heat
Modular	Only the cables you need; better airflow
Redundant	Two supplies in servers; one fails, the other carries the load; hot swappable

11 Printers

LASER STEP	WHAT HAPPENS
1 Processing	The page is rasterized into a bitmap in the printer's memory
2 Charging	The primary charge roller (or corona wire) puts a uniform negative charge (about minus 600 V) on the drum
3 Exposing	The laser writes the image, removing charge where toner should stick
4 Developing	Negatively charged toner from the developer roller clings to the exposed areas
5 Transferring	The transfer roller (or corona) gives the paper a positive charge that pulls toner off the drum
6 Fusing	Heat (about 180 C) and pressure melt the toner into the paper; the fuser is hot

LASER STEP	WHAT HAPPENS
7 Cleaning	A blade scrapes leftover toner off the drum; a lamp erases the charge

PRINTER	KEY PARTS	MAINTENANCE
Laser	Imaging drum, toner cartridge, fuser, transfer belt or roller, pickup rollers, separation pad, duplexing assembly	Replace toner; apply the maintenance kit (fuser, rollers, pads) at the page count; calibrate; clean with a toner vacuum, never compressed air
Inkjet	Ink cartridges, print head, carriage belt, feed rollers, duplexer	Clean and align print heads; replace cartridges; calibrate; clear jams; keep heads from drying
Thermal	Heating element, feed assembly, special heat-sensitive paper, no ink	Replace paper; clean the heating element with isopropyl alcohol; remove debris; keep paper away from heat
Impact (dot matrix)	Print head with pins, ribbon, tractor feed, multipart carbonless paper	Replace ribbon, print head, and paper; the only type that prints multipart forms
3D	Filament (**FDM**) or resin, print bed, nozzle	Level the bed; clear the nozzle; store filament dry

Deployment and settings

- Unbox, remove shipping tape and locks, place near power and network away from heat and dust; update firmware; install the right driver for the **OS**
- **PCL**: fast, printer-dependent, everyday output. PostScript: device-independent page description, exact and consistent, used in graphics and publishing
- Connectivity: **USB** (one **PC**), Ethernet, wireless; sharing: a printer shared from a **PC** versus a print server that manages queues and drivers centrally
- Settings: duplex (both sides), orientation (portrait, landscape), tray selection and paper size, quality and draft mode
- Security: user authentication, badge release, audit logs, secured (held) prints released with a **PIN** at the device
- Scan services: scan to email (**SMTP** settings), scan to folder (**SMB** share and credentials), scan to cloud; **ADF** for stacks, flatbed for books and cards

12 Mobile devices

LAPTOP PART	REPLACEMENT NOTES
Battery	Lithium-ion; internal on most modern laptops; discharge and disconnect before other work; swollen means replace now
Keyboard and keys	Ribbon cable under the palm rest; individual keycaps on scissor or butterfly mechanisms
RAM	**SODIMM**; many thin laptops solder it (not upgradable)
Storage	2.5-inch **SATA** or **M.2** (**SATA** or **NVMe**); check the key and length
Wireless card	**M.2** (2230) card; the two antenna leads (main and aux) route through the hinge to the lid; black and white or gray leads must go to their marked terminals
Camera and microphone	In the lid bezel; cable through the hinge; a physical privacy shutter on some
Biometrics and NFC	Fingerprint reader in the power button or palm rest; smart card or **NFC** reader for badges

Connections and accessories

- **USB-C** (reversible, data, video, power), microUSB and miniUSB (older), Lightning (Apple, older); **NFC** tap; Bluetooth pairing; tethering and hotspot share a phone's cellular data
- Docking station: a full set of ports and power with one connection (Thunderbolt or **USB-C**, or proprietary). Port replicator: adds ports only, usually **USB**
- Accessories: stylus (active pens need pairing or a battery), headsets, speakers, webcams; trackpad, drawing pad, track point

- Cellular: **3G**, **4G LTE**, **5G**; **SIM** card or **eSIM** profile identifies the subscriber; airplane mode; data roaming
- Bluetooth pairing: enable Bluetooth, enable pairing (discoverable), find the device, enter the **PIN**, test the connection
- Location: **GPS** satellites (needs sky) and cellular or **Wi-Fi** positioning (works indoors, less precise)
- **MDM**: corporate versus **BYOD** device configurations, policy enforcement (passcodes, encryption, remote wipe), corporate application deployment
- Synchronization: mail, calendar, contacts, cloud storage, business apps; watch data caps on cellular; sync over **Wi-Fi** when possible

13 Virtualization and cloud

TERM	MEANING
Type 1 hypervisor	Bare metal, runs directly on hardware (server virtualization); best performance
Type 2 hypervisor	Runs as an application on a host **OS** (desktop virtualization, testing)
Sandbox	An isolated **VM** to open suspicious files or test changes safely
Test and development	Many **OS** versions on one machine; snapshots to roll back
Application virtualization	Run legacy software or another **OS**'s programs; cross-platform virtualization
VDI	Virtual desktop infrastructure: desktops run on servers, users connect with thin clients
Containers	Share the host kernel; lighter than **VMs**; package an application with its dependencies
VM requirements	**CPU** virtualization enabled, enough **RAM** and storage for each guest, network mode (bridged, **NAT**, host-only), and security (patch guests, isolate, guard against **VM** escape)

CLOUD MODEL	MEANING
Public	Provider's shared infrastructure, pay as you go
Private	Dedicated to one organization, on premises or hosted
Hybrid	Private plus public, workloads move between them
Community	Shared by organizations with common needs (government, healthcare)
IaaS	Rent virtual machines, storage, and networks; you manage the **OS** up
PaaS	A platform to build and run applications; the provider manages **OS** and runtime
SaaS	A finished application over the web; email, office suites, **CRM**

Cloud characteristics

- Shared resources (multitenancy, cheaper) versus dedicated resources (isolation, predictable performance)
- Metered utilization: pay for what you use; ingress (data in) is usually free, egress (data out) is billed
- Elasticity: capacity grows and shrinks with demand automatically; availability: uptime commitments through redundancy
- File synchronization: the same files on every device and in the cloud; conflicts and deleted-file propagation are the traps

14 Troubleshooting

The method (CompTIA's best practice)

- 1 Identify the problem: gather information, question the user, identify changes, back up before making changes. 2 Establish a theory of probable cause; question the obvious; research. 3 Test the theory. 4 Establish a plan of action and implement it. 5 Verify full functionality and implement preventive measures. 6 Document findings, actions, and outcomes.

BOARD, RAM, CPU, POWER SYMPTOM	LIKELY CAUSE AND FIRST CHECK
No power, no fans	Outlet, cord, **PSU** switch and voltage selector, **PSU** (test with a tester or paperclip), front panel header
Fans spin, no POST, beeps	Beep codes point to **RAM** or video; reseat **RAM** and card; one module at a time
Blank screen after POST	Monitor input, cable, **GPU** seating and power; onboard video
Random shutdown, overheating, burning smell	Dust, failed fan, dried thermal paste, failing **PSU**; check temperatures in firmware
Proprietary crash screen (BSOD, pinwheel)	Driver, **RAM** (run a memory test), storage, overheating; note the stop code
Sluggish performance, application crashes	**RAM** shortage, thermal throttling, failing drive, malware
Unusual noise	Fan bearing (whine), **HDD** (clicking, grinding), coil whine under load
Capacitor swelling or leaking	Board or **PSU** failure; replace
Inaccurate date and time	**CMOS** battery (**CR2032**)

DRIVE AND RAID SYMPTOM	LIKELY CAUSE AND FIRST CHECK
Clicking, grinding	**HDD** mechanical failure; back up immediately, replace
Bootable device not found	Boot order, loose cable, dead drive, corrupted boot record, **USB** stick left in
S.M.A.R.T. failure warning	Drive predicts its own failure; back up and replace
Extended read and write times, low IOPS	Failing drive, fragmentation on **HDD**, nearly full **SSD**, wrong **SATA** mode
Data loss or corruption	Failing drive, bad cable, power loss during writes; run chkdsk after a backup
RAID failure, array missing, audible alarm	A failed member; replace the drive and rebuild; check the controller and its battery; a missing array may be a controller or driver
Missing drives in the OS	Not initialized or partitioned, driver, cable, power, disabled port in firmware
LED status indicators	Amber or red on a drive bay means degraded or failed

VIDEO AND DISPLAY SYMPTOM	LIKELY CAUSE AND FIRST CHECK
No image, incorrect input source	Select the right input; check the cable and adapter; try another cable
Dim image	Brightness, power saving, failing backlight or inverter; shine a flashlight at the screen
Fuzzy image, sizing issues	Non-native resolution, scaling, **VGA** over a long cable, projector focus
Flashing or flickering screen	Loose cable, refresh rate, driver, failing panel or inverter
Incorrect color, distorted image	Bent **VGA** pins, cable, driver, color profile, degaussing on **CRTs**
Dead pixels	Panel defect; warranty policy; stuck pixels may recover
Burn-in	Static image on **OLED** or plasma; pixel shift and screensavers prevent it
Burnt-out bulb, intermittent projector shutdown	Replace the lamp (hours counter); overheating from a clogged filter
Audio issues over HDMI or DisplayPort	Select the display as the output device; cable supports audio

MOBILE SYMPTOM	LIKELY CAUSE AND FIRST CHECK
Poor battery health, short life	Aged battery (check health percentage), background apps, brightness, radios; replace
Swollen battery	Stop using it; do not puncture; replace and dispose properly
Improper charging	Cable, adapter wattage, dirty or damaged port, wrong charger
Overheating	Charging while gaming, direct sun, malware, a failing battery

MOBILE SYMPTOM	LIKELY CAUSE AND FIRST CHECK
Liquid damage	Power off, do not charge, dry thoroughly; indicators inside show exposure
Digitizer issues, cursor drift, touch calibration	Cracked digitizer, screen protector, calibration utility, moisture
Poor or no connectivity	Airplane mode, **Wi-Fi** or cellular toggles, **SIM** seating, forget and rejoin the network, carrier settings
Malware, cannot install apps, degraded performance	Unofficial app sources, full storage, **OS** out of date, factory reset as last resort
Stylus does not work	Battery or pairing on active pens, wrong pen model, digitizer
Broken screen, damaged port	Replace the assembly; ports are soldered on many devices

NETWORK SYMPTOM	LIKELY CAUSE AND FIRST CHECK
Intermittent wireless	Weak signal, interference, channel overlap, roaming between **APs**, driver
Slow speeds	Congestion, wrong cable category, duplex mismatch, bandwidth hogs, a hub
Limited connectivity, APIPA address	No **DHCP**: link, cable, **VLAN**, **DHCP** server
Intermittent internet	**ISP**, modem, router overheating, **DNS**; test the modem directly
Jitter, high latency, poor VoIP	Congestion; no QoS; wireless for voice; satellite links
Port flapping	Link going up and down: bad cable or connector, failing **NIC**, switch port, duplex or speed mismatch
External interference	Microwaves, motors, fluorescent ballasts, neighbors' **Wi-Fi**; move to 5 **GHz** or shielded cable
Authentication failures	Wrong password or key, expired certificate, **RADIUS** server, **MAC** filtering, time skew

PRINTER SYMPTOM	LIKELY CAUSE AND FIRST CHECK
Lines or streaks down the page	Dirty or scratched drum, low toner, dirty print head (inkjet), a scratch on the fuser
Faded prints	Low toner or ink, economy mode, transfer roller worn, humidity
Speckling	Loose toner in the printer; leaking cartridge; clean
Double or echo images (ghosting)	Drum not cleaned, worn drum, fuser problem
Toner not fused, smears when touched	Fuser failed or not reaching temperature; wrong paper
Garbled print	Wrong driver or language (**PCL** versus PostScript), bad cable, corrupted job, low printer memory
Paper jams, multipage misfeed	Worn pickup rollers or separation pad, damp or wrong paper, overfilled tray, debris
Paper not feeding, tray not recognized	Rollers, tray seating, paper guides, tray sensor, firmware
Multiple prints pending, frozen queue	Restart the print spooler service, clear the queue, check the printer status and connectivity
Incorrect page orientation or size	Driver settings, application settings, tray paper size mismatch
Grinding noise	Gears, rollers, drum drive, foreign object; maintenance kit
Finishing issues (staple jams, hole punch)	Clear the finisher, refill staples, check the finisher tray sensors
Connectivity issues	**IP** change on **DHCP** (use a reservation), **Wi-Fi**, cable, sleep mode, firewall

15 Networking and hardware tools

TOOL	USE
Crimper	Attaches **RJ45** or **RJ11** plugs to cable ends

TOOL	USE
Cable stripper	Removes the jacket without nicking conductors
Punchdown tool	Seats wires into a 110 or 66 block or a keystone jack and trims them
Cable tester	Checks wire map, opens, shorts, split pairs; certifiers measure to category
Toner probe (tone generator and probe)	Finds the far end of a cable in a bundle or closet
Loopback plug	Tests a **NIC** or switch port by looping transmit to receive
Wi-Fi analyzer	Shows signal strength, channels, and interference; site surveys
Network tap	Copies traffic on a link for a protocol analyzer without disrupting it
Multimeter	Voltage, continuity, resistance; **PSU** rails and cables
Power supply tester	Checks each **PSU** rail and the power-good signal
POST card	Reads the firmware's progress codes when a board will not boot
ESD strap and mat	Prevents static damage while handling boards, **RAM**, and cards

16 Acronyms

Every acronym used in this course, 272 in all. In lessons and on this sheet, hover over an acronym for the short version; click it, or click one below, for the longer explanation.

ACRONYM	STANDS FOR	MEANING
12VHPWR	12 volt high power connector	The 16-pin connector on new graphics cards rated up to 600 W.
3G	Third-generation cellular	The older mobile data generation, being switched off by carriers.
4G	Fourth-generation cellular	The LTE generation of mobile broadband.
5G	Fifth-generation cellular	The current mobile network generation with higher speed and lower latency than LTE.
802.11	IEEE 802.11	The family of Wi-Fi standards.
802.3af	IEEE 802.3af	The original Power over Ethernet standard: about 15.4 W at the switch, 12.95 W at the device.
802.3at	IEEE 802.3at	PoE+: about 30 W at the switch, 25.5 W at the device.
802.3bt	IEEE 802.3bt	PoE++: 60 W (Type 3) or up to 90 W (Type 4) using all four pairs.
AAA	Authentication, authorization, and accounting	Prove who you are, decide what you may do, record what you did.
AAAA	Quad-A record	The DNS record that maps a name to an IPv6 address.
AC	Alternating current	Mains power, which reverses direction 60 times a second in North America.
ACK	Acknowledgment	The TCP flag or message confirming data arrived.
ADF	Automatic document feeder	The tray on a scanner or printer that feeds a stack of pages.
AES	Advanced Encryption Standard	The standard symmetric cipher, with 128, 192, or 256-bit keys.
AHCI	Advanced Host Controller Interface	The SATA controller mode that enables native command queuing and hot plug.
AIO	All-in-one	A liquid CPU cooler with a sealed pump and radiator, or a computer built into its monitor.
AM4	AMD socket AM4	AMD's PGA socket for Ryzen processors through 2022.
AM5	AMD socket AM5	AMD's current LGA socket for Ryzen processors with DDR5.
AMD	Advanced Micro Devices	The other major maker of x86 CPUs and of Radeon graphics.
AMD-V	AMD Virtualization	AMD's hardware support for running virtual machines.
AP	Access point	The radio that bridges Wi-Fi clients onto the wired network.
API	Application programming interface	The defined way one program talks to another, such as the web endpoints a cloud service exposes.
APIPA	Automatic Private IP Addressing	The 169.254.x.x address a host gives itself when it cannot reach a DHCP server.
ARM	Advanced RISC Machine	The low-power processor design in phones, tablets, and newer laptops.
ARP	Address Resolution Protocol	Finds the MAC address that goes with a known IPv4 address on the local segment.
ATX	Advanced Technology Extended	The standard full-size desktop motherboard and power supply form factor, 12 x 9.6 inches.
B+M	B and M keyed M.2 module	An M.2 SSD notched for both B and M key slots, usually a SATA drive.
BIOS	Basic Input/Output System	The original PC firmware: 16-bit, text setup, MBR disks up to 2 TB.
BNC	Bayonet Neill-Concelman connector	The twist-lock coax connector used on older LANs, test equipment, and CCTV.

ACRONYM	STANDS FOR	MEANING
BSOD	Blue screen of death	The Windows stop error screen from a driver or hardware fault.
BYOD	Bring your own device	Staff use personal phones and laptops for work.
CAS	Column address strobe (latency)	The delay in clock cycles before memory returns requested data; lower is faster.
CCFL	Cold cathode fluorescent lamp	The backlight in older LCDs, driven by a high-voltage inverter.
CCTV	Closed-circuit television	Security cameras on a private video system.
CD	Compact disc	Optical media holding about 700 MB.
CEC	Consumer Electronics Control	The HDMI feature that lets one remote control several devices.
CF	CompactFlash card	A larger, older flash memory card used in professional cameras and industrial gear.
CIFS	Common Internet File System	An older name for the SMB file-sharing protocol.
CL16	CAS latency 16	A memory module that answers in 16 clock cycles.
CMOS	Complementary metal-oxide semiconductor	The small battery-backed memory that keeps firmware settings and the clock.
CNAME	Canonical name record	A DNS alias: one name points to another name.
CPU	Central processing unit	The processor that executes instructions.
CR2032	CR2032 coin cell	The 3 V lithium coin battery that keeps CMOS settings and the clock.
CRM	Customer relationship management	Software for tracking customers and sales, usually SaaS.
CRT	Cathode ray tube	The old glass-tube monitor; holds dangerous voltage even when unplugged.
CSM	Compatibility Support Module	UEFI's legacy mode that boots BIOS-style operating systems.
DaaS	Desktop as a service	VDI hosted by a cloud provider.
DB9	D-subminiature 9-pin connector	The 9-pin serial port connector.
dBm	Decibels relative to one milliwatt	An absolute power level: 0 dBm is 1 mW; Wi-Fi signals are negative numbers such as -65 dBm.
DC	Direct current	Current in one direction: batteries, power supply outputs, and PoE.
DCI-P3	Digital Cinema Initiatives P3 color space	A wide color gamut used for video and film work.
DDR	Double data rate (memory)	The family of SDRAM generations (DDR3, DDR4, DDR5) that transfer twice per clock cycle.
DDR3	Double data rate 3 SDRAM	The memory generation before DDR4, 240-pin DIMMs at 1.5 V.
DDR4	Double data rate 4 SDRAM	The common memory generation: 288-pin DIMM, 260-pin SODIMM, 1.2 V, 2133 to 3200 MT/s.
DDR4-2666	DDR4 at 2666 megatransfers per second	A slower DDR4 speed rating.
DDR4-3200	DDR4 at 3200 megatransfers per second	A DDR4 module speed rating.
DDR5	Double data rate 5 SDRAM	The current memory generation: higher speed, 1.1 V, on-module voltage regulation.
DE-15	D-subminiature E shell, 15 pins	The formal name of the VGA connector.
DFS	Dynamic frequency selection	Wi-Fi channels in the 5 GHz band that must yield to radar.
DHCP	Dynamic Host Configuration Protocol	Hands out IP addresses, masks, gateways, and DNS servers automatically (UDP 67 and 68).
DIMM	Dual in-line memory module	The full-size desktop memory stick.
DKIM	DomainKeys Identified Mail	Outgoing mail is digitally signed; the public key is published in DNS.
DMARC	Domain-based Message Authentication, Reporting, and Conformance	A DNS policy telling receivers what to do with mail that fails SPF or DKIM, and where to send reports.
DNS	Domain Name System	Translates names such as www.example.com into IP addresses (UDP 53, TCP 53 for zone transfers).
DOCSIS	Data Over Cable Service Interface Specification	The standard cable modems use to carry internet over the TV coax.
DORA	Discover, Offer, Request, Acknowledge	The four-step DHCP lease exchange.
DP	DisplayPort	The computer-oriented digital video connector with daisy-chaining and high bandwidth.
DRAM	Dynamic random-access memory	The main system memory technology, refreshed constantly.
DSL	Digital subscriber line	Internet over the telephone line through an RJ11 jack; speed falls with distance from the exchange.
DSSS	Direct-sequence spread spectrum	The spreading technique used by early 2.4 GHz Wi-Fi (802.11b).
DVD	Digital versatile disc	Optical media holding 4.7 GB, or 8.5 GB dual layer.
DVI	Digital Visual Interface	The older white digital video connector; DVI-D digital, DVI-A analog, DVI-I both.
DVI-A	DVI analog	A DVI connector that carries only an analog VGA-style signal.
DVI-D	DVI digital	A DVI connector that carries only a digital signal.
DVI-I	DVI integrated	A DVI connector carrying both digital and analog signals.
ECC	Error-correcting code memory	Memory that detects and fixes single-bit errors; used in servers and workstations.
EPS	Entry-level Power Supply specification (CPU power connector)	The 4-pin or 8-pin 12 V connector near the CPU socket.
eSATA	External SATA	A SATA port on the outside of the case for external drives.
ESD	Electrostatic discharge	Static electricity that can destroy chips; prevented with a wrist strap, mat, and antistatic bags.
eSIM	Embedded SIM	A SIM built into the phone and provisioned by software.
FDM	Fused deposition modeling	3D printing that melts plastic filament through a nozzle layer by layer.

ACRONYM	STANDS FOR	MEANING
FTP	File Transfer Protocol	Plaintext file transfer on TCP 21 (control) and 20 (data); replace with SFTP or FTPS.
FTPS	FTP Secure	FTP wrapped in TLS, ports 990 and 989 (implicit) or 21 with an upgrade (explicit).
GB	Gigabyte	About a billion bytes; the unit for RAM, storage, and data caps.
GG45	GG45 connector	A backward-compatible Cat 7 and Cat 8 connector that also accepts RJ45 plugs.
GHz	Gigahertz	Billions of cycles per second; CPU clock speed and Wi-Fi band frequency.
GPS	Global Positioning System	Satellite positioning, also used as a stratum 0 time reference.
GPT	GUID Partition Table	The modern disk partition scheme: any disk size, up to 128 partitions, required for UEFI boot.
GPU	Graphics processing unit	The processor that renders images; on a card or built into the CPU.
HD	High definition	Video resolutions of 1280 x 720 (HD) and 1920 x 1080 (Full HD).
HDD	Hard disk drive	Spinning magnetic storage: cheap per terabyte, slower, and fragile.
HDMI	High-Definition Multimedia Interface	The common digital video and audio connector for monitors and TVs.
HEPA	High-efficiency particulate air (filter)	A filter that traps fine dust; used in vacuums safe for electronics and toner.
HR	Human resources	The department that handles staff records; its documents are the exam's example of sensitive printouts.
HSM	Hardware security module	A tamper-resistant appliance or card that generates, stores, and uses cryptographic keys.
HTTP	Hypertext Transfer Protocol	Web traffic in the clear on TCP 80.
HTTPS	Hypertext Transfer Protocol Secure	HTTP inside TLS on TCP 443.
HVAC	Heating, ventilation, and air conditioning	The building's climate systems, which keep server rooms cool and are themselves networked controls.
Hz	Hertz	Cycles per second; display refresh rate, AC mains frequency, and signal frequency.
IaaS	Infrastructure as a service	The provider supplies virtual machines, storage, and networks; you manage the operating system and up.
ID	Identifier or identification	A name or number that uniquely labels a user, device, session, or record.
IDE	Integrated Drive Electronics	The legacy parallel ATA drive interface, and the legacy SATA controller mode.
IDS	Intrusion detection system	Watches a copy of the traffic and alerts on attacks; cannot block them.
IMAP	Internet Message Access Protocol	Reads mail that stays on the server, synced across devices; TCP 143, secure 993.
IOPS	Input/output operations per second	A measure of how many small reads and writes a drive handles per second.
IoT	Internet of Things	Networked devices such as cameras, sensors, thermostats, and smart TVs.
IP	Internet Protocol	The Layer 3 protocol that addresses and routes packets between networks.
IPS	In-plane switching	The LCD panel type with the best color accuracy and viewing angles.
IPv4	Internet Protocol version 4	32-bit addresses written as four decimal octets, such as 192.168.1.10.
IPv6	Internet Protocol version 6	128-bit addresses in eight hex groups; no broadcast, built-in autoconfiguration.
iSCSI	Internet SCSI	SCSI storage commands over a TCP/IP network, giving block-level storage without Fibre Channel.
ISO	International Organization for Standardization; also an ISO disc image	The standards body, and the .iso file format that holds a complete disc image.
ISP	Internet service provider	The company that connects you to the internet.
IT	Information technology	Computers, networks, and the people who run them.
ITX	Information Technology eXtended (Mini-ITX)	The 6.7-inch square small-form-factor motherboard size.
LAN	Local area network	One site's network: a building, floor, or campus that one organization owns and runs.
LC	Lucent connector	The small square snap-in fiber connector, the most common on modern transceivers.
LCD	Liquid crystal display	The flat panel that shapes light from a backlight into an image; TN, IPS, and VA are the panel types.
LDAP	Lightweight Directory Access Protocol	Queries a directory such as Active Directory for users and groups; TCP 389, secure 636.
LDAPS	LDAP Secure	LDAP over TLS on TCP 636.
LED	Light-emitting diode	A small light source: the status lights on a port, the lamp in cheap fiber transmitters, and modern display backlights.
LGA	Land grid array	A CPU socket where the pins are in the socket and the chip has flat contacts (Intel, and AMD AM5).
LTE	Long-Term Evolution	4G cellular data; the workhorse mobile broadband standard.
M.2	M.2 form factor	The small card slot for SSDs and wireless cards, keyed B, M, or B+M.
MAC	Media access control (address)	The 48-bit hardware address burned into a network interface, used at Layer 2.
macOS	Mac operating system	Apple's desktop operating system.
MAN	Metropolitan area network	A city-sized network: bigger than a LAN, smaller than a WAN.
MB	Megabyte	About a million bytes; file sizes and older media capacities.
MBR	Master boot record	The legacy partition scheme: disks up to 2 TB, four primary partitions, BIOS boot.
MDIX	Medium-dependent interface crossover	Auto-MDIX ports sense the cable and swap transmit and receive pairs, so crossover cables are rarely needed.

ACRONYM	STANDS FOR	MEANING
MDM	Mobile device management	Software that enrolls phones and tablets and enforces policy: passcodes, encryption, apps, remote wipe.
MHz	Megahertz	Millions of cycles per second; memory speed, radio channels, and older CPU clocks.
microSD	Micro Secure Digital card	The fingernail-sized flash card in phones, cameras, and single-board computers.
MIMO	Multiple input, multiple output	Using several antennas at once to send parallel data streams for more speed.
ms	Millisecond	One thousandth of a second; latency and response times.
mSATA	Mini-SATA	A small SATA SSD card used in older laptops before M.2.
MT/s	Megatransfers per second	The speed rating of memory: millions of data transfers per second.
MU-MIMO	Multi-user MIMO	The access point serves several clients at the same time with separate streams.
MX	Mail exchanger record	The DNS record that says which server accepts email for a domain.
NAS	Network-attached storage	A file server appliance that shares folders over the regular network using SMB or NFS.
NAT	Network address translation	Rewrites private inside addresses to a public address at the router so many hosts can share it.
NAT64	Network address translation 6 to 4	Lets IPv6-only hosts reach IPv4-only servers by translating between the two.
NFC	Near-field communication	Very short range radio (about 4 cm) at 13.56 MHz for tap-to-pay and badge readers.
NFS	Network File System	The Unix and Linux file-sharing protocol.
NIC	Network interface card	The adapter that connects a computer to the network, wired or wireless.
NS	Name server record	The DNS record listing the authoritative name servers for a zone.
NTP	Network Time Protocol	Synchronizes clocks across the network (UDP 123); wrong time breaks Kerberos, certificates, and logs.
NVMe	Non-Volatile Memory Express	The fast SSD interface that talks to PCIe lanes directly, usually in an M.2 slot.
OFDMA	Orthogonal frequency-division multiple access	Wi-Fi 6 splits a channel into small sub-channels so many clients share it efficiently.
OLED	Organic light-emitting diode	A display whose pixels emit their own light: perfect blacks, thin, but prone to burn-in.
OM3	Optical multimode 3	Laser-optimized 50 micron multimode fiber, aqua, 10 Gbps to about 300 m.
ONT	Optical network terminal	The box that converts the fiber provider's light signal to Ethernet at the customer premises.
OS	Operating system	The software that manages hardware and runs applications: Windows, macOS, Linux, iOS, Android.
OSI	Open Systems Interconnection model	The seven-layer reference model used to describe where a protocol or device does its work.
PaaS	Platform as a service	The provider runs the platform (OS, runtime, database); you deploy code.
PAN	Personal area network	A network within arm's reach, such as Bluetooth between a phone and earbuds.
PC	Personal computer	A desktop or laptop for one user.
PC4-25600	PC4-25600 bandwidth label	The DDR4-3200 speed written as 25,600 MB/s of bandwidth.
PCI	Peripheral Component Interconnect	The older expansion bus; PCIe (PCI Express) is its modern serial successor.
PCL	Printer Command Language	HP's page description language, fast and common in office printing.
PDF	Portable Document Format	Adobe's fixed-layout document file.
PGA	Pin grid array	A CPU package with pins on the chip that drop into holes in the socket (AMD AM4).
PIN	Personal identification number	A short numeric secret, a something-you-know factor.
PJL	Printer Job Language	Commands that wrap a print job to set options and switch languages.
PoE	Power over Ethernet	Delivers DC power to devices over the data cable: 802.3af 15.4 W, 802.3at 30 W, 802.3bt 60 to 90 W.
POP3	Post Office Protocol version 3	Downloads mail to one device and usually deletes it from the server; TCP 110, secure 995.
POST	Power-on self-test	The firmware's hardware check at power-up; failures show as beep codes or a POST card reading.
PPI	Pixels per inch	Pixel density; higher is sharper.
PRL	Preferred roaming list	The carrier's list of networks a phone may use, updated over the air.
PS	PostScript	Adobe's page description language, favored for graphics and publishing.
PSU	Power supply unit	Converts wall AC to the DC voltages a computer needs: 12 V, 5 V, 3.3 V.
PTR	Pointer record	The reverse DNS record: IP address back to a name.
PVC	Polyvinyl chloride	The ordinary cable jacket, which gives off toxic smoke; plenum cable is required in air spaces.
PWM	Pulse-width modulation	Controls fan speed (4-pin headers) and LED brightness by switching power on and off rapidly.
PXE	Preboot Execution Environment	Booting a computer from the network to install or image an operating system.
QHD	Quad high definition	2560 x 1440 resolution.
QR	Quick response code	The square barcode a phone camera reads.
RADIUS	Remote Authentication Dial-In User Service	The AAA server behind Wi-Fi Enterprise, VPNs, and 802.1X; UDP 1812 and 1813.
RAID	Redundant array of independent disks	Combining disks for speed, capacity, or fault tolerance: 0 stripe, 1 mirror, 5 stripe with parity, 6 double parity, 10 mirrored stripes.
RAM	Random-access memory	The working memory that holds running programs and data; lost at power off.
RDP	Remote Desktop Protocol	Microsoft's remote graphical desktop on TCP 3389.

ACRONYM	STANDS FOR	MEANING
RFID	Radio-frequency identification	Tags read by radio for badges, inventory, and asset tracking.
RG-59	Radio Guide 59 coaxial cable	Thinner coax for short runs such as CCTV, often with a BNC connector.
RG-6	Radio Guide 6 coaxial cable	The modern coax grade for cable TV and cable modems, with an F-type connector.
RGB	Red, green, blue	The three color channels of displays and the name of decorative lighting headers.
RISC	Reduced instruction set computer	A processor design with a small set of simple instructions, as in ARM.
RJ11	Registered jack 11	The small telephone plug, used for analog phone lines and DSL.
RJ45	Registered jack 45	The eight-pin plug on Ethernet twisted-pair cables (properly an 8P8C connector).
ROM	Read-only memory	Non-volatile memory holding firmware.
RPM	Revolutions per minute	Rotation speed: hard drive platters at 5400 or 7200, fans, and laboratory centrifuges.
RS-232	Recommended Standard 232 serial	The classic serial port: slow, simple, and still used for console access and legacy devices.
RX	Receive	The receiving side of a link or the receiving fiber strand.
S.M.A.R.T	Self-Monitoring, Analysis, and Reporting Technology	Drive self-diagnostics that warn of impending failure.
SaaS	Software as a service	Complete applications delivered over the web, such as email and office suites.
SAN	Storage area network	A dedicated network that gives servers block-level access to shared disk arrays.
SAS	Serial Attached SCSI	The enterprise drive interface for servers; faster, dual-ported, and accepts SATA drives.
SATA	Serial ATA	The standard drive interface: SATA III at 6 Gb/s, 7-pin data and 15-pin power cables.
SC	Subscriber connector	The larger square push-pull fiber connector, common on older gear and patch panels.
SCADA	Supervisory control and data acquisition	Systems that monitor and control industrial processes such as power, water, and manufacturing.
SCP	Secure Copy Protocol	Simple file copying over SSH, TCP 22.
SCSI	Small Computer System Interface	The old parallel drive and peripheral bus; SAS and iSCSI carry its command set today.
SD	Secure Digital card	The flash memory card used in cameras and devices; miniSD and microSD are the smaller sizes.
SFP	Small form-factor pluggable transceiver	The hot-swappable module that gives a switch port its fiber or copper interface, 1 Gbps.
SFTP	SSH File Transfer Protocol	File transfer inside an SSH session, TCP 22.
SFX	Small form factor power supply	A compact power supply for small cases.
SIM	Subscriber identity module	The card or embedded chip that identifies a phone to the cellular network.
SLA	Service level agreement	The contract stating measurable service commitments, such as 99.9% uptime.
SLAAC	Stateless address autoconfiguration	IPv6 hosts build their own address from the router's advertised prefix, no DHCP server needed.
SMB	Server Message Block	Windows file and printer sharing on TCP 445.
SMT	Simultaneous multithreading	One physical core runs two threads at once (Intel calls it Hyper-Threading).
SMTP	Simple Mail Transfer Protocol	Sends email between servers on TCP 25; clients submit on 587 with TLS.
SNMP	Simple Network Management Protocol	Polls and receives alerts from network devices (UDP 161 polls, 162 traps).
SODIMM	Small outline DIMM	The short memory module used in laptops and small PCs.
SOHO	Small office, home office	A small network with a consumer-grade router doing NAT, DHCP, Wi-Fi, and firewall in one box.
SPAN	Switched port analyzer	A switch feature that mirrors traffic from ports or VLANs to a monitoring port.
SPF	Sender Policy Framework	A DNS TXT record listing which servers may send mail for a domain.
SQL	Structured Query Language	The language used to query databases; also the name of the servers on TCP 1433 (Microsoft) and 3306 (MySQL).
SRAM	Static random-access memory	Faster memory that needs no refresh, used for CPU cache.
SSD	Solid-state drive	Flash storage with no moving parts: fast, silent, and shock resistant.
SSH	Secure Shell	Encrypted remote command-line access and file transfer on TCP 22; replaces Telnet and FTP.
SSID	Service set identifier	The Wi-Fi network name that access points broadcast and clients join.
ST	Straight tip connector	The round bayonet twist-lock fiber connector; legacy.
STP	Spanning Tree Protocol	Blocks redundant switch links so frames cannot loop, while keeping them ready as backups.
SYN	Synchronize flag	The first packet of a TCP handshake.
Syslog	System logging protocol	Sends device log messages to a central server on UDP 514 with severity levels 0 to 7.
T568A	TIA/EIA 568A wiring standard	One of the two pin-outs for RJ45 cables; green pair on pins 1 and 2.
T568B	TIA/EIA 568B wiring standard	The more common RJ45 pin-out; orange pair on pins 1 and 2.
TACACS+	Terminal Access Controller Access-Control System Plus	Cisco's AAA protocol for device administration on TCP 49; encrypts the whole payload and separates authentication from authorization.
TB	Terabyte	About a trillion bytes; the unit for large drives.
TCP	Transmission Control Protocol	Reliable, ordered, connection-based transport: three-way handshake, acknowledgments, retransmission.
TFTP	Trivial File Transfer Protocol	Bare-bones UDP 69 file transfer with no login, used for firmware and config files.
TFX	Thin form factor power supply	A slim power supply for low-profile desktop cases.

ACRONYM	STANDS FOR	MEANING
TLD	Top-level domain	The last part of a domain name, such as .com or .org.
TLS	Transport Layer Security	The encryption layer under HTTPS and most secure protocols; use 1.2 or 1.3.
TN	Twisted nematic	The fastest and cheapest LCD panel type, with poor viewing angles and color.
TPM	Trusted Platform Module	A security chip on the motherboard that stores keys and measurements; required for BitLocker and Windows 11.
TRIM	TRIM command	Tells an SSD which blocks are no longer in use so it can erase them ahead of time.
TTL	Time to live	A counter that keeps packets and DNS records from living forever.
TV	Television	A display or broadcast receiver; smart TVs are IoT devices.
TX	Transmit	The sending side of a link or the sending fiber strand.
TXT	Text record	A free-text DNS record, used for SPF, DKIM, DMARC, and domain ownership checks.
UDP	User Datagram Protocol	Connectionless, fast transport with no delivery guarantee; used for voice, video, DNS, DHCP.
UEFI	Unified Extensible Firmware Interface	Modern firmware that replaces BIOS: GPT disks, Secure Boot, mouse-driven setup.
UHD	Ultra high definition	3840 x 2160 resolution, commonly called 4K.
UPS	Uninterruptible power supply	A battery unit that keeps equipment running through a power cut and conditions the power.
US	United States	The country; US standards, 60 Hz mains, and US connector conventions.
USB	Universal Serial Bus	The standard port for peripherals, storage, and charging.
USB-A	USB Type-A connector	The classic flat rectangular USB plug.
USB-C	USB Type-C connector	The small reversible connector that carries USB, DisplayPort, Thunderbolt, and power delivery.
USB4	Universal Serial Bus 4	The 40 Gbps USB generation over USB-C, based on Thunderbolt.
UTM	Unified threat management	One appliance bundling firewall, IDS/IPS, antivirus, content filtering, and VPN.
UTP	Unshielded twisted pair	The standard copper network cable: four twisted pairs, no shield.
VA	Vertical alignment	An LCD panel type with high contrast and deep blacks, slower than IPS.
VAC	Volts AC	An alternating-current voltage rating, such as 120 VAC.
VCSEL	Vertical-cavity surface-emitting laser	The cheap laser used in multimode fiber transceivers.
VDI	Virtual desktop infrastructure	User desktops run as virtual machines in the data center; the user connects with a thin client.
VGA	Video Graphics Array	The blue 15-pin analog video connector.
VLAN	Virtual local area network	A logical LAN on a switch: ports in different VLANs are separate broadcast domains.
VM	Virtual machine	A complete computer running as software on a hypervisor.
VNC	Virtual Network Computing	A cross-platform remote desktop protocol.
VoIP	Voice over IP	Telephone calls carried as packets on the data network.
VPN	Virtual private network	An encrypted tunnel across an untrusted network such as the internet.
VRAM	Video RAM	The memory on a graphics card that holds frames and textures.
VT	Virtualization technology	Intel's hardware virtualization support, enabled in firmware.
WAN	Wide area network	Connects LANs across cities or countries, usually over links leased from a provider.
Wi-Fi	Wireless fidelity	The trade name for IEEE 802.11 wireless networking.
WISP	Wireless internet service provider	An ISP that delivers internet by fixed wireless from a tower to an antenna on the building.
WLAN	Wireless local area network	A LAN delivered over Wi-Fi instead of cables.
WPA2	Wi-Fi Protected Access 2	Wi-Fi security using AES with CCMP; still common and acceptable.
WPA3	Wi-Fi Protected Access 3	The current Wi-Fi security standard: SAE in Personal mode, stronger encryption, protected management frames.
WPS	Wi-Fi Protected Setup	The push-button or PIN method for joining Wi-Fi; its PIN mode is insecure and should be disabled.
ZIF	Zero insertion force	A socket or connector that clamps without pushing, used for CPUs and laptop ribbon cables.